

**UNIVERSITY NAME***probability and statistics — Final Exam**Year: 2021***Exercise 1:**

A parking lot has 10 spaces numbered from 1 to 10. The number of cars  $N$  arriving at the lot in one hour follows a Poisson distribution with parameter  $\lambda > 0$ . It is assumed that drivers choose a space at random and independently of one another.

Let  $G$  be the event “the driver chooses space number 1” and  $X$  be the random variable “number of drivers who choose space number 1”.

Calculate:

1.  $E(N)$
2.  $P(G)$
3.  $P(X = k \mid N = m)$  (distinguish the cases  $k > m$  and  $k \leq m$ )
4.  $P(X = k)$
5.  $E(X)$  and  $\text{Var}(X)$
6. The probability generating function of  $N$  and that of  $X$ .

**Answer Area****Exercise 2:**

Let  $X$  be a uniformly distributed random variable on the interval  $[-1, 1]$ .

1. Find the distribution function of the random variable  $Y = 4 - X^2$ .
2. Compute  $P(2 \leq Y \leq 3.5)$ .

**Answer Area**